

## Multiple Choice (10 marks)

1. Facial flushing is seen in approximately 50% of East Asians and is thought to be due to a deficiency of?
  - (a) ALDH 1
  - (b) ALDH 2
  - (c) ALDH 3
  - (d) ALDH 4
2. Which of the following is not a zinc-dependent enzyme?
  - (a) Superoxide dismutase
  - (b) Alkaline phosphatase
  - (c) Glutathione peroxidase
  - (d) DNA polymerase
3. Ulcerative Colitis
  - (a) Carries a risk of colonic cancer
  - (b) Can be treated with a LOFFLEX diet
  - (c) Can be caused by small intestinal bacterial overgrowth
  - (d) Can be caused by milk intolerance
4. A deficiency of which vitamin has been associated with enamel defects and increased risk of dental caries?
  - (a) Vitamin A
  - (b) Folic acid
  - (c) Vitamin C
  - (d) Vitamin D
5. Which of the following statements is not true?
  - (a) Vegan diets are likely to be deficient in protein
  - (b) Cereals are deficient in lysine
  - (c) Legumes are deficient in methionine
  - (d) Dairy products are rich source of tryptophan

## True / False (10 marks)

*Answer True or False*

6. True or False: Niacin serves as a coenzyme in reduction reactions during fatty acid synthesis.
7. True or False: Sodium is a dietary component that has been shown to improve hypertension in humans.
8. True or False: Metformin is the first-line medication recommended for patients with type 2 diabetes and obesity as of 2020.
9. True or False: Epidemiological measures of effect assess the strength of the reversibility between an exposure and an outcome.
10. True or False: Starch mainly contains amylose, which is a polymer of glucose with alpha 1,4-linkages.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the role of ATP as the immediate source of chemical energy in skeletal muscle tissue and explain how it facilitates muscle contraction during physical activity.
12. Discuss the effectiveness of various treatments for Binge Eating Disorder (BED) and analyze why benzodiazepines, such as Diazepam and Alprazolam, are not considered effective options.

*End of Paper*